

Calvin Lynch

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ISTQB Certified Software Test Engineer, First Class Honours Software Development graduate, with practical experience in test automation (C#, Reqnroll, BDD), CI/CD pipeline integration, and AI-assisted QA tooling using Claude on AWS Bedrock. Background spans regulated software, DevOps, and full-stack observability. Looking to move into a dedicated automation engineering role.

EDUCATION

BSc (Hons) Software Development — First Class Honours 2021 – 2025
Technological University of the Shannon Midlands Midwest (TUS), Limerick | QQI Level 8 | GPA: 73.66%

WORK EXPERIENCE

Graduate Software Test Engineer — Release & Regression Team *July 2025 – Present*
Kneat Solutions

- Automated critical API regression scenarios in C# using Reqnroll (BDD), implementing feature-file scenarios covering expected product behavior and customer-reported defects across current and previous application versions, while applying SOLID principles to build maintainable and reusable test automation executed nightly in CI.
- Co-developed an automated test reporting platform using Python, Poetry, Azure DevOps REST APIs, and CI/CD pipelines. Integrated Claude-based AI agents via AWS Bedrock to analyze test plans, validate compliance, generate customer-ready reports, and support requirements traceability using SpecSync.
- Co-developed as part of a team an AI-assisted PR security analysis platform using Claude, OWASP Top 10, and CVE-based vulnerability detection — awarded 1st place out of 16 teams in an internal AI Engineering competition. Enabled faster PR reviews and automated senior Security Manager for escalation based on vulnerability severity.
- Execute regression, exploratory, and PR testing; logging and validating defects using Azure DevOps and Git across multiple application modules. API testing, validating HTTP responses (status codes, headers, payload structure) and functional behavior of REST endpoints against expected business logic.

Software Engineer Intern – Platform Engineering *Jan 2024 – Aug 2024*
Progress Software Corporation (8 months)

- Embedded within a Platform Operations team gaining direct experience with DevOps practices, CI/CD pipelines, automation, and Agile delivery (SAFe framework).
- Built a cloud cost analysis dashboard integrating data from Azure CLI, CassandraDB, and Prometheus, visualized in Grafana for internal use — end-to-end ownership from data pipeline to dashboard.
- Implemented MELT observability (Metrics, Events, Logs, Traces) using Prometheus for metric scraping and alerting across platform services.
- Deployed a Jaeger instance for distributed tracing and service dependency analysis.
- Daily stack: Python, Docker, Kubernetes (K3s), GitHub Actions, FastAPI, Flask, CassandraDB, InfluxDB, Grafana, Argo CD.
- Participated in sprint reviews, PI planning, code reviews, and retrospectives.

TECHNICAL SKILLS

Containers & Orchestration	Docker, Docker Compose, Kubernetes (K3s), Argo CD, Traefik
CI/CD & Automation	GitHub, GitHub Actions, Azure DevOps, CI/CD pipeline design
Observability	Prometheus, Grafana, Alertmanager, Node Exporter, PromQL, MELT, Jaeger
Scripting & Backend	Linux, Python, Bash
Databases	InfluxDB, PostgreSQL, CassandraDB, MySQL
Tooling & AI	Git, GitHub, Azure DevOps, Swagger, SpecSync, Claude, AWS Bedrock, OpenRouter
Languages	Python, Java, C#, Javascript, HTML, CSS
Frameworks & Libraries	Fast Api, React, Spring Boot

CERTIFICATIONS & AWARDS

ISTQB Certified Tester — Foundation Level	ISTQB	2026
Junior Penetration Tester	TryHackMe	2025
5 - Year Service Award	Absolute Hotel Limerick	2023
QQI Level 5 — Software Development (Distinctions in all modules)	LCFE	2021

PROJECTS

Full portfolio: <https://cv-calvin.vercel.app/projects>

Air Quality Monitoring System — Final Year Project

- Designed and built a portable air quality monitoring system using a Raspberry Pi 5 as the edge computing device, integrating environmental sensors to collect real-time PM2.5, PM10, and CO2 data.
- Processed all sensor data locally on-device using Python, applying edge computing principles to reduce reliance on cloud processing and minimize latency.
- Exposed collected data via a FastAPI REST API, with time-series data persisted in InfluxDB hosted on AWS, enabling scalable cloud-backed storage and querying.
- Containerized all backend services using Docker for consistent, portable deployments across development and production environments.
- Integrated Prometheus and Grafana for infrastructure-level monitoring and observability of the full deployed stack, including alerting configuration.
- Built a React web application and a SwiftiOS mobile app to provide real-time data visualization, giving users cross-platform access to live and historical air quality readings.

Monitoring & Alerting Stack

- Designed and self-hosted a full observability stack on a Raspberry Pi using Docker, covering metrics collection, dashboard visualization, and automated alerting end-to-end.
- Configured Prometheus with custom scrape jobs targeting system services; wrote PromQL queries to evaluate real-time performance metrics and define alert thresholds.
- Deployed Node Exporter to expose host-level system metrics (CPU, memory, disk, network) for collection by Prometheus.
- Built Grafana dashboards to visualise live and historical metrics, providing clear system health overviews across all monitored services.
- Configured Alertmanager with custom alerting rules and webhook integrations to route notifications on service failure; validated the full alerting pipeline through deliberately simulated downtime scenarios.

Task Management System

2024

- Designed and developed a secure Task Management System using Java Spring Boot, exposing RESTful APIs for full CRUD functionality across task resources.
- Integrated Spring Security with JWT-based authentication, OAuth2 support (Google login), and OTP-based multi-factor authentication to provide layered access control.
- Configured HTTPS using self-signed TLS certificates to secure all API communication and documented all endpoints using Swagger for developer usability.
- Implemented a comprehensive automated test suite using JUnit and Mockito, integrated into a GitHub Actions CI/CD pipeline with JaCoCo for code coverage tracking and reporting.